Strength and Deformation of Rockfill Material Based on Large-Scale Triaxial Compression Tests. II: Influence of Particle Breakage

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Abstract: The amount of particle breakage of Tacheng rockfill material (TRM) in the process of shearing was quantitatively measured by a relative breakage index. The fractal dimension of TRM at the final state of the tests was found to be linearly related to the void ratio and logarithm of the confining pressure. The influences of the relative breakage index on the friction angle, modulus, and deformation were investigated based on a series of large-scale triaxial compression tests on TRM. An increase in the relative breakage index leads to an increase in the volumetric strain and to decreases in the peak state friction angle, critical state friction angle, deviatoric strain, and critical state void ratio. The test results were used to propose simple formulations pertaining to the relative breakage index for the strength and deformation indexes (e.g., peak state friction angle, critical state friction angle, initial elastic modulus, secant modulus at 50% of the peak strength, volumetric strain, deviatoric strain, and critical state void ratio). **DOI: 10.1061/(ASCE)GT.1943-5606.0001177.** © *2014 American Society of Civil Engineers.*

Author keywords: Rockfill material; Fractal dimension; Relative breakage index; Critical state; Void ratio; Confining pressure; Friction angle; Modulus.

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Introduction

For a crushable granular soil, particle breakage considerably influences the soil's strength behavior (Marsal 1967; Hardin 1985; Indraratna et al. 1993; Lade et al. 1996; McDowell et al. 1996; Einav 2007a, b). It is essential to propose a proper breakage index that can comprehensively represent the degree of particle breakage during loadings. Particle breakage indexes (Lee and Farhoomand 1967; Marsal 1967; Hardin 1985; Lade et al. 1996) are obtained based on particle-size distributions before and after a test. Miura et al. (2003) used increments of fines content (75 μm or smaller) as the breakage factor during the consolidation and shearing process. The increase in the particle surface area was also used to quantify the degree of particle crushing in the loading process (Miura and Yamamoto 1976; Miura and O-Hara 1979). Other methods for defining the breakage index include a relative particle breakage index based on fractal theory (Einav 2007a). Muir Wood and Maeda (2008) proposed a grading state index similar to that proposed by Einav (2007a) to quantify the degree of particle breakage. Constitutive models, for example elastoplastic models (Salim and Indraratna 2004; Daouadji and Hicher 2010), bounding surface models (Russell and Khalili 2004), and disturbed state concept models (Varadarajan et al. 2003, 2006), were developed by incorporating the breakage index to capture the stress-strain behaviors of crushable granular soils.

The mechanical behaviors of rockfill materials have mainly been investigated via large-scale triaxial tests (Marsal 1967; Leps 1970; Marschi et al. 1972; Charles and Watts 1980; Barton and Kjaernsli 1981; Varadarajan et al. 1997; Indraratna et al. 1998; Varadarajan et al. 1999; Gupta 2000; Varadarajan et al. 2003, 2006; Anderson and Fair 2008; Gupta 2009a, b; Chu et al. 2010; Honkanadavar 2010; Honkanadavar and Gupta 2010; Seif El Dine et al. 2010; Honkanadavar et al. 2011; Araei et al. 2012; Honkanadavar et al. 2012; Sevi and Ge 2012; Vasistha et al. 2012; Xu et al. 2012; Vasistha et al. 2013; Xiao et al. 2014a, b), multiaxial tests (Desai et al. 1982; Desai and Siriwardane 1983; Janardhanam and Desai 1983; Desai and Faruque 1984; Desai and Salami 1987; Desai et al. 1995; Desai and Toth 1996; Desai 2001; Desai and El-Hoseiny 2005), and true triaxial tests (Shi 2008; Xiao et al. 2011a, b, 2012). These tests indicated that particle breakage influences the strength of rockfill materials considerably (Marsal 1967; Indraratna et al. 1993, 1998; Gupta 2000; Varadarajan et al. 2002, 2003, 2006; Gupta 2009a; Abbas 2011).

McDowell et al. (1996) proposed a conceptual model for the evolution of a granular medium under stress. McDowell and Bolton (1998) found that the smaller particles generated by fracture are in geometrically self-similar configurations under increasing stress. The simulation of particle breakage was linked to the energy required for the creation of new surfaces (McDowell and Bolton 1998). In addition, the discrete element method (DEM) is an effective tool for investigating the crushing behaviors of granular soils under high stresses and with various stress paths (Cheng et al. 2003, 2004; Lobo-Guerrero and Vallejo 2005, 2006a, b; Lobo-Guerrero et al. 2006; Wang and Yan 2012; Lashkari and Golchin 2014).

The state-dependent behaviors of Tacheng rockfill material (TRM) can be described using a state-dependent constitutive model (Been and Jefferies 1985; Gudehus 1996; Li 1997; Li and Wang 1998; Gajo and Muir Wood 1999; Wan and Guo 1999; Li and Dafalias 2000, 2002; Wang et al. 2002; Yang and Li 2004; Yao et al. 2004; Lashkari 2009; Loukidis

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and Salgado 2009). A companion paper (Xiao et al. 2014b) discusses the influence of density and pressure on the strength and deformation characteristics of TRM through a series of large-scale triaxial compression tests. This paper mainly focuses on the influence of particle breakage on the strength and deformation behaviors of TRM.

The relative breakage index proposed by Einav (2007a) is adopted for TRM to describe the amount of particle crushing that occurs during the shearing process. Furthermore, this paper discusses the influence of particle breakage on the peak state friction angle, critical state friction angles, initial elastic modulus, secant modulus at 50% of the peak strength, volumetric and deviatoric strains, and critical state void ratio of TRM.

Fractal Dimension

Following the breakage-mechanic theory proposed by Einav (2007a), the initial grading (IG), current grading (CG), and ultimate grading (UG) for TRM, as shown in Fig. 1, can be, respectively, expressed as follows:

$$F_0(d) = \left(\frac{d}{d_M}\right)^{3-\alpha_0} \tag{1a}$$

$$F_c(d) = \left(\frac{d}{d_M}\right)^{3-\alpha_c} \tag{1b}$$

$$F_u(d) = \left(\frac{d}{d_M}\right)^{3-\alpha_u} \tag{1c}$$

where $F_0(d)$, $F_c(d)$, and $F_u(d)$ = formulations of the IG, CG, and UG, respectively; α_0, α_c, and α_u = fractal dimensions of the IG, CG, and UG, respectively; d = diameter; and d_M = maximum diameter.

The current fractal dimension α_c in this paper is obtained by using Eq. (1*b*) to simulate the grading at the end of the tests. The current fractal dimension α_c is varied with the initial confining pressure at $e_0 = 0.189$, as shown in Fig. 2(a), and with the initial void ratio at $p_0 = 1.6$ MPa, as shown in Fig. 2(b). Fig. 2 illustrates that the simulations based on Eq. (1*b*) agree well with the gradations of TRM at the end of the tests. Table 1 presents the values of the fractal dimension α_c for TRM at various initial confining pressures and void ratios. The values of the fractal dimension (as listed in Table 1) could be used to determine the values of relative breakage index (as listed in Table 2). An increase in the initial confining pressure leads to an increase in the fractal dimension of TRM at a given initial void ratio, as shown in Fig. 3(a). In contrast, an increase in the initial void ratio leads to a decrease in the fractal dimension of TRM at a given initial confining pressure, as shown in Fig. 3(b). The gradient of the fractal dimension α_c with the logarithm of the initial confining pressure p_0 remains unchanged, as shown in Fig. 3(a). Furthermore, the gradient of the fractal dimension α_c with the initial void ratio e_0 is independent of the initial confining pressure p_0, as shown in Fig. 3(b). As shown in Fig. 3, the initial fractal dimension α_0 pertaining to the initial confining pressure p_0 and initial void ratio e_0 can be expressed as

$$\alpha = \alpha_{e0} - \chi_\alpha e_0 + k_\alpha \lg\frac{p_0}{p_a} \tag{2}$$

where α_{e0}, χ_α, and k_α = material parameters. Based on the test data in Fig. 3 and Eq. (2), the material parameters (i.e., α_{e0}, χ_α, k_α) are determined by a least-squares analysis. The values of the material

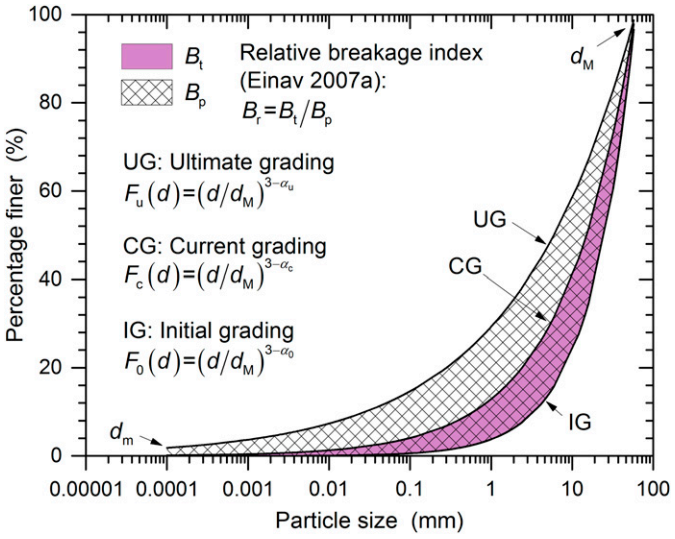


Fig. 1. Definition of the relative breakage index

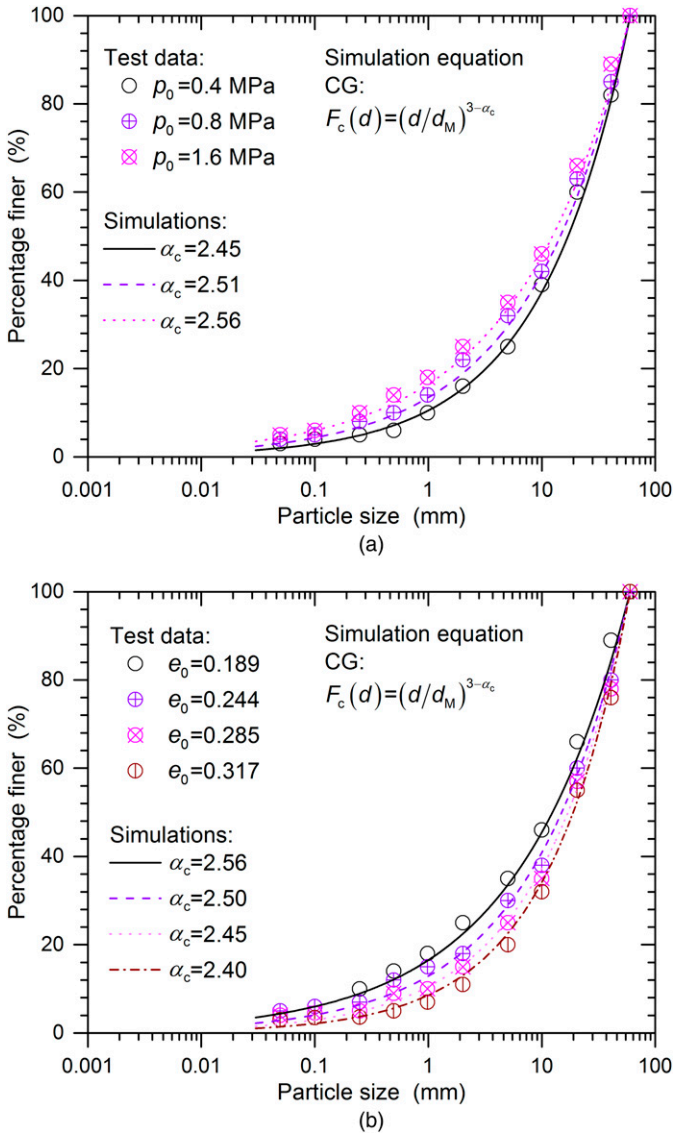


Fig. 2. Variation in the particle-size distributions of TRM after shearing: (a) initial confining pressure at $e_0 = 0.189$; (b) initial void ratio at $p_0 = 1.6$ MPa

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Table 1. Fractal Dimension of TRM after Shearing

Initial confining pressure (p_0) (MPa)	Initial void ratio (e_0)	Current fractal dimension (α_c)[a]
0.4	0.189	2.45
	0.244	2.39
	0.285	2.35
	0.317	2.33
0.8	0.189	2.51
	0.244	2.45
	0.285	2.4
	0.317	2.35
1.6	0.189	2.56
	0.244	2.50
	0.285	2.45
	0.317	2.40

[a]Fractal dimension defined by McDowell et al. (1996) and Einav (2007a).

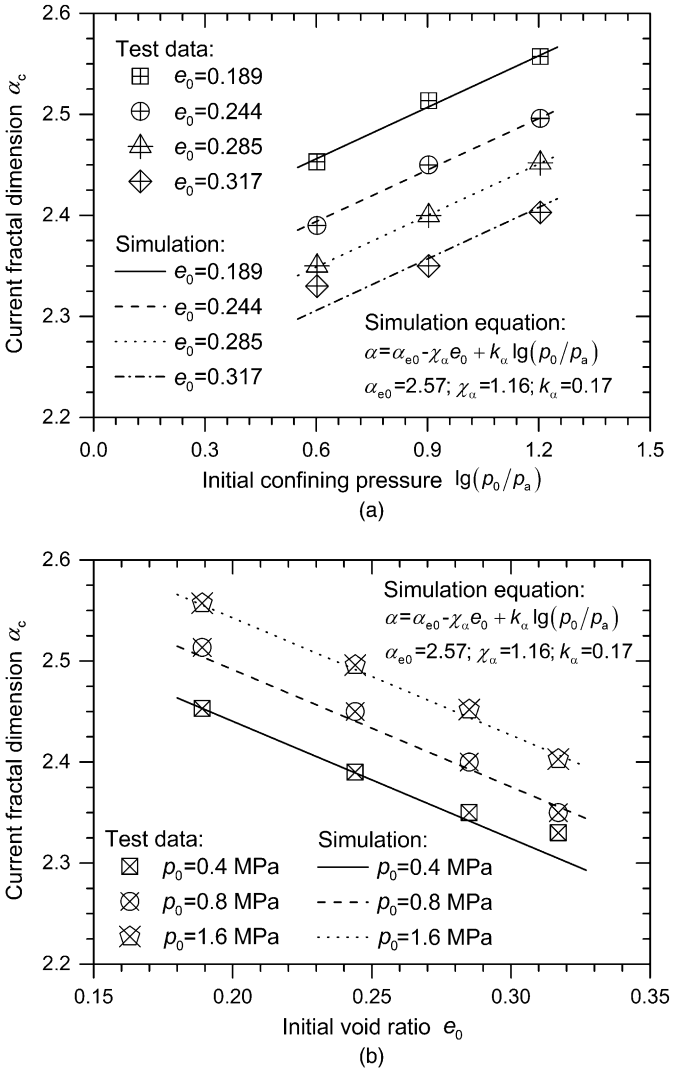


Fig. 3. Variations in the fractal dimension of TRM: (a) initial confining pressure; (b) initial void ratio

parameters are listed in Table 3. Fig. 3 illustrates that the simulations based on Eq. (2) agree well with the test data of TRM at different densities and pressures.

The initial fractal dimension α_0 is regarded as 2.1 [see Fig. 2(a) from Xiao et al. 2014b]. The maximum value of the fractal dimension α_c, as presented in Table 1, is 2.56 at $p_0 = 1.6$ MPa and $e_0 = 0.189$. The ultimate fractal dimension α_u should be greater than 2.56. The ultimate fractal dimension α_u for rockfill materials should be 2.7 based on data from the literature (Yang and Juo 2001).

Definition of the Relative Breakage Index

Particle breakage indexes have been proposed by many researchers (Marsal 1967; Lee and Farhoomand 1967; Hardin 1985; Lade et al. 1996; Lackenby et al. 2007) based on particle-size distributions before and after a test. The relative breakage index proposed by Hardin (1985) is regarded as one of the most effective tools for quantifying the degree of fragmentation for crushable aggregates. According to the definition of the relative breakage index provided by Hardin (1985), it is possible for all of the particles in a sample of soil to be crushed such that no particles with diameters greater than 0.074 mm are present under extremely high stresses. This assumption is invalid, as noted by Einav (2007a). The notion that the grain-size distribution tends to be fractal is becoming increasingly accepted, which is approved by Einav (2007a). As shown in Fig. 1, based on the work by Einav (2007a), the relative breakage index can be expressed as

$$B_r = \frac{B_t}{B_p} \tag{3}$$

where B_r = relative breakage index; B_p = breakage potential; and B_t = total breakage potential (Hardin 1985).

As shown in Fig. 1, according to the definition by Einav (2007a), the breakage potential B_p can be obtained by integrating the area over the $\log d$ scale as

$$B_p = \int_{d_m}^{d_M} \left[F_u(d) - F_0(d)\right] d(\log d) \tag{4}$$

where d_m = minimum diameter of the sieving size. The total breakage potential B_t can be calculated as

$$B_t = \int_{d_m}^{d_M} \left[F_c(d) - F_0(d)\right] d(\log d) \tag{5}$$

Substitution of Eqs. (4) and (5) into Eq. (3) yields

$$B_r = \frac{\int_{d_m}^{d_M} \left[F_u(d) - F_0(d)\right] d(\log d)}{\int_{d_m}^{d_M} \left[F_c(d) - F_0(d)\right] d(\log d)} = \frac{\int_{d_m}^{d_M} \left[F_u(d) - F_0(d)\right] d^{-1} dd}{\int_{d_m}^{d_M} \left[F_c(d) - F_0(d)\right] d^{-1} dd} = \frac{(\alpha - \alpha_0)(3 - \alpha_u)}{(\alpha_u - \alpha_0)(3 - \alpha)} \tag{6}$$

Eq. (6) can be used to determine the relative breakage index of TRM.

Substitution of Eq. (2) into Eq. (6) yields

$$B_r = \left(\frac{3 - \alpha_u}{\alpha_u - \alpha_0}\right)\left(\frac{\alpha_{e0} - \alpha_0 - \chi_\alpha e_0 + k_\alpha \lg\frac{p_0}{p_a}}{3 - \alpha_{e0} + \chi_\alpha e_0 - k_\alpha \lg\frac{p_0}{p_a}}\right) \tag{7}$$

Eq. (7) provides the relative breakage index of TRM based on the initial confining pressure p_0 and initial void ratio e_0. The degree of particle crushing at the end of the tests can be estimated by Eq. (7).

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The values of the relative breakage index B_r can be calculated by Eq. (6) based on the values of α_c for TRM in Table 1. The values of the relative breakage index B_r of TRM at the end of the tests are listed in Table 2. Fig. 4 presents the variations in the relative breakage index of TRM with the initial void ratio and initial confining pressure. The relative breakage index of TRM at a given initial void ratio increases with an increase in the initial confining pressure, as shown in Fig. 4(a). The B_r of TRM at a given initial confining pressure increases with a decrease in the initial void ratio, as shown in Fig. 4(b). The simulations based on Eq. (7) are in good agreement with the test data on the relative breakage index of TRM at different confining pressures and void ratios, as shown in Fig. 4.

Table 2. Relative Breakage Index of TRM

Initial confining pressure (p_0) (MPa)	Initial void ratio (e_0)	Relative breakage index (B_r) (%)[a]
0.4	0.189	32.27
	0.244	23.77
	0.285	19.23
	0.317	17.16
0.8	0.189	42.50
	0.244	31.82
	0.285	25.00
	0.317	19.23
1.6	0.189	51.62
	0.244	39.29
	0.285	32.12
	0.317	25.38

[a]Relative breakage index defined by Einav (2007a).

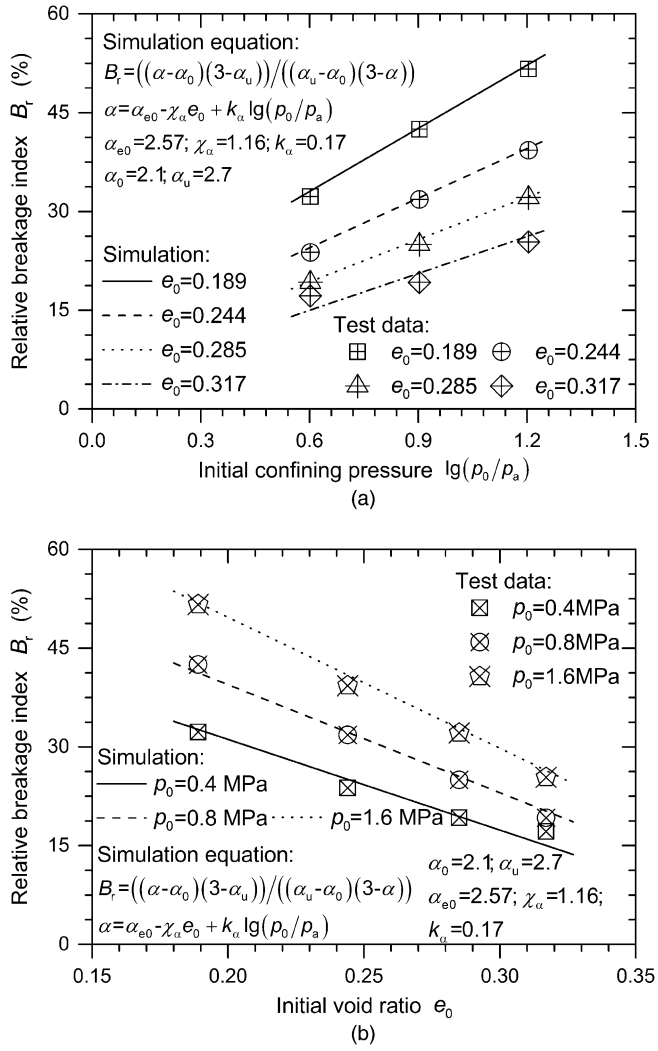


Fig. 4. Variations in the relative breakage index of TRM: (a) initial confining pressure; (b) initial void ratio

Influence of the Relative Breakage Index on the Friction Angle

Fig. 5 illustrates the influence of the relative breakage index B_r on the peak state friction angle ϕ_{ps} and the critical state friction angle ϕ_{cs}. An increase in the relative breakage index B_r leads to a decrease in the peak state friction angle ϕ_{ps} and critical state friction angle ϕ_{cs}. The peak state friction angle ϕ_{ps} and critical state friction angle ϕ_{cs} can be linearly expressed as functions of the breakage index B_r as

$$\phi_{ps} = \phi_{ps0} - k_{ps}B_r/100 \tag{8a}$$

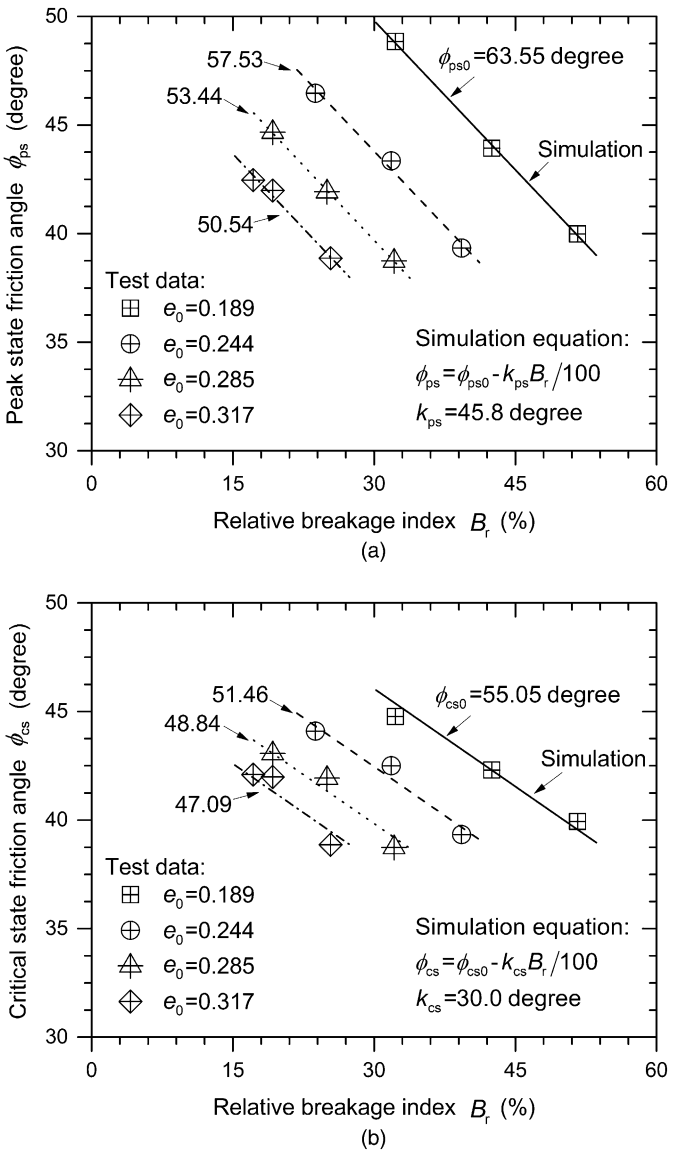


Fig. 5. Influence of the relative breakage index of TRM: (a) peak state friction angle; (b) critical state friction angle

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$$\phi_{cs} = \phi_{cs0} - k_{cs}B_r/100 \tag{8b}$$

where ϕ_{ps0}, k_{ps}, ϕ_{cs0}, and k_{cs} = material parameters, which can be determined by a least-squares analysis. The values of these material parameters (i.e., ϕ_{ps0}, k_{ps}, ϕ_{cs0}, k_{cs}) are listed in Table 3. Fig. 5 illustrates that the values of k_{ps} and k_{cs} remain constant, whereas ϕ_{ps0} and ϕ_{cs0} are dependent on the initial void ratio e_0. As shown in Fig. 6, the value of ϕ_{ps0} and ϕ_{cs0} decrease with an increase in the initial void ratio e_0. ϕ_{ps0} and ϕ_{ps0} can be expressed as linear formulations of e_0 as

$$\phi_{ps0} = \phi_{eps} - \chi_{eps}e_0 \tag{9a}$$

$$\phi_{cs0} = \phi_{ecs} - \chi_{ecs}e_0 \tag{9b}$$

where ϕ_{eps}, χ_{eps}, ϕ_{ecs}, and χ_{ecs} = material parameters, which can be determined by a least-squares analysis. The values of these material parameters (i.e., ϕ_{eps}, χ_{eps}, ϕ_{ecs}, χ_{ecs}) are listed in Table 3. Substituting Eq. (9) into Eq. (8) yields

Table 3. Values of the Material Constants for the Breakage Behaviors of TRM

Equation	Symbol	Value
Eq. (2)	α_{e0}	2.57
	χ_{α}	1.16
	k_{α}	0.17
Eq. (6)	α_0	2.1
	α_u	2.7
Eq. (8a)	ϕ_{ps0}	63.55° for $e_0 = 0.189$
		57.53° for $e_0 = 0.244$
		53.44° for $e_0 = 0.285$
		50.54° for $e_0 = 0.317$
	k_{ps}	45.80°
Eq. (8b)	ϕ_{cs0}	55.05° for $e_0 = 0.189$
		51.46° for $e_0 = 0.244$
		48.84° for $e_0 = 0.285$
		47.09° for $e_0 = 0.317$
	k_{cs}	30.00°
Eq. (9a)	ϕ_{eps}	82.65°
	χ_{eps}	1.02°
Eq. (9b)	ϕ_{ecs}	66.81°
	χ_{ecs}	0.63°
Eq. (12)	E_{B0}	196.12 MPa
	k_{E0}	22.06 MPa
Eq. (13)	E_{B50}	250.12 MPa for $p_0 = 0.4$ and 0.8 MPa
		445.98 MPa for $p_0 = 1.6$ MPa
	k_{E50}	20 MPa for $p_0 = 0.4$ and 0.8 MPa
		20 MPa for $p_0 = 1.6$ MPa
Eq. (14a)	ε_{v0}	16.58% for $e_0 = 0.189$
		11.34% for $e_0 = 0.244$
		7.94% for $e_0 = 0.285$
		4.81% for $e_0 = 0.317$
	$k_{\varepsilon v}$	0.35
Eq. (14b)	ε_{s0}	20.46% for $e_0 = 0.189$
		18.70% for $e_0 = 0.244$
		17.64% for $e_0 = 0.285$
		16.57% for $e_0 = 0.317$
	$k_{\varepsilon s}$	0.12
Eq. (15a)	$\varepsilon_{\varepsilon v0}$	33.75%
	$\chi_{\varepsilon v0}$	0.91
Eq. (15b)	$\varepsilon_{\varepsilon s0}$	26.10%
	$\chi_{\varepsilon s0}$	0.30
Eq. (18)	e_{B0}	0.37
	k_e	0.40

$$\phi_{ps} = \phi_{eps} - \chi_{eps}e_0 - k_{ps}B_r/100 \tag{10a}$$

$$\phi_{cs} = \phi_{ecs} - \chi_{ecs}e_0 - k_{cs}B_r/100 \tag{10b}$$

Eq. (10) provides estimations of the peak state friction angle ϕ_{ps} and critical state friction angle ϕ_{cs} based on the initial void ratio e_0 and breakage index B_r.

The increases in the peak state friction angle ϕ_{rps} and critical state friction angle ϕ_{rcs} caused by the particle crushing can be expressed as

$$\phi_{rps} = \phi_{ps0} - \phi_{ps} = k_{ps}B_r/100 \tag{11a}$$

$$\phi_{rcs} = \phi_{cs0} - \phi_{cs} = k_{cs}B_r/100 \tag{11b}$$

Fig. 7 shows that ϕ_{rps} and ϕ_{rcs} can be linearly related to the breakage index B_r. In addition, the predictions by Eq. (11) agree well with the test results. ϕ_{rps} and ϕ_{rcs} are independent of the initial void ratio e_0 for a given relative breakage index B_r.

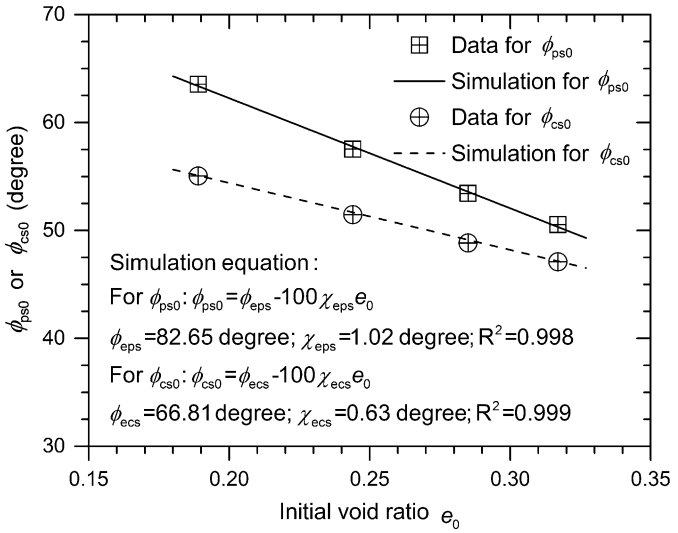


Fig. 6. Relationship between ϕ_{ps0}, ϕ_{cs0}, and e_0 for TRM

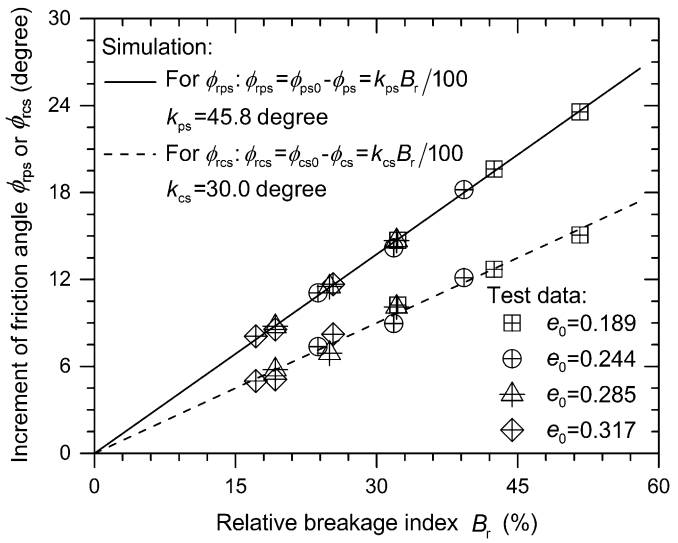


Fig. 7. Relationship between the increment of friction angles and relative breakage index for TRM

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Influence of the Relative Breakage Index on the Modulus

Fig. 8 illustrates the influence of the relative breakage index B_r on the initial elastic modulus E_{0i} of TRM at different initial void ratios. The relationship between E_{0i} and B_r for TRM can be linearly expressed as

$$E_{0i} = k_{E0}B_r - E_{B0} \quad (12)$$

where k_{E0} and E_{B0} = material parameters, which can be determined by a least-squares analysis. The values of the two material parameters (i.e., k_{E0} and E_{B0}) are listed in Table 3. Eq. (12) is a suitable method for mapping between E_{0i} and B_r. In addition, the relationship between E_{0i} and B_r is independent of the initial void ratio e_0, as shown in Fig. 8.

Fig. 9 illustrates the influence of the relative breakage index B_r on the secant modulus E_{50} at 50% of the peak strength. The relationship between E_{50} and B_r for TRM can be expressed as

$$E_{50} = k_{E50}B_r - E_{B50} \quad (13)$$

where k_{E50} and E_{B50} = material parameters, which can be determined by a least-squares analysis. Table 3 lists the values of the two material parameters (i.e., k_{E50} and E_{B50}). The relationship between E_{50} and B_r is dependent on the initial confining pressure p_0. The gradient k_{E50} in Eq. (13) is independent of the confining pressure, whereas the value of E_{B50} at a lower confining pressure (e.g., $p_0 = 0.4$ or 0.8 MPa) differs from that at a higher confining pressure (e.g., $p_0 = 1.6$ MPa). The predictions based on Eq. (13) agree well with the test results, as shown in Fig. 9.

Influence of the Relative Breakage Index on Deformation

Fig. 10 presents the influence of the relative breakage index B_r on the volumetric strain ε_v and deviatoric strain ε_s at the end of the tests. The

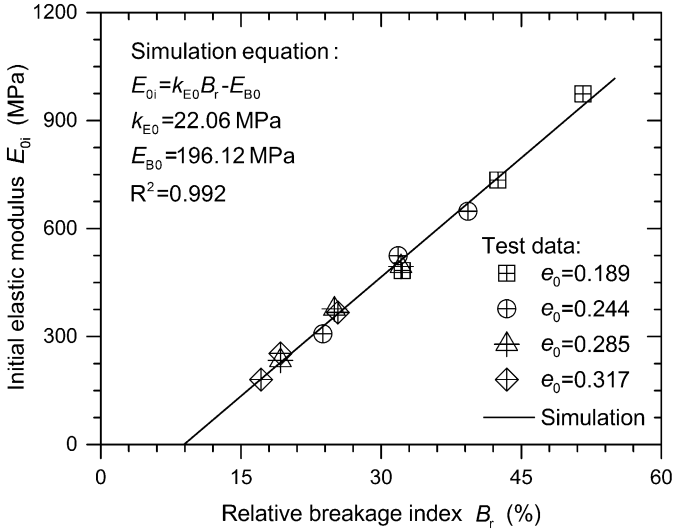


Fig. 8. Relationship between the initial elastic modulus and relative breakage index for TRM

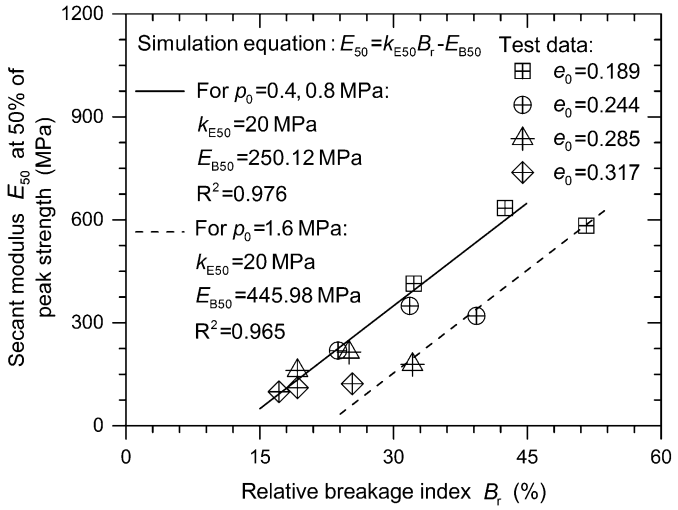


Fig. 9. Relationship between the secant modulus at 50% of the peak strength and relative breakage index for TRM

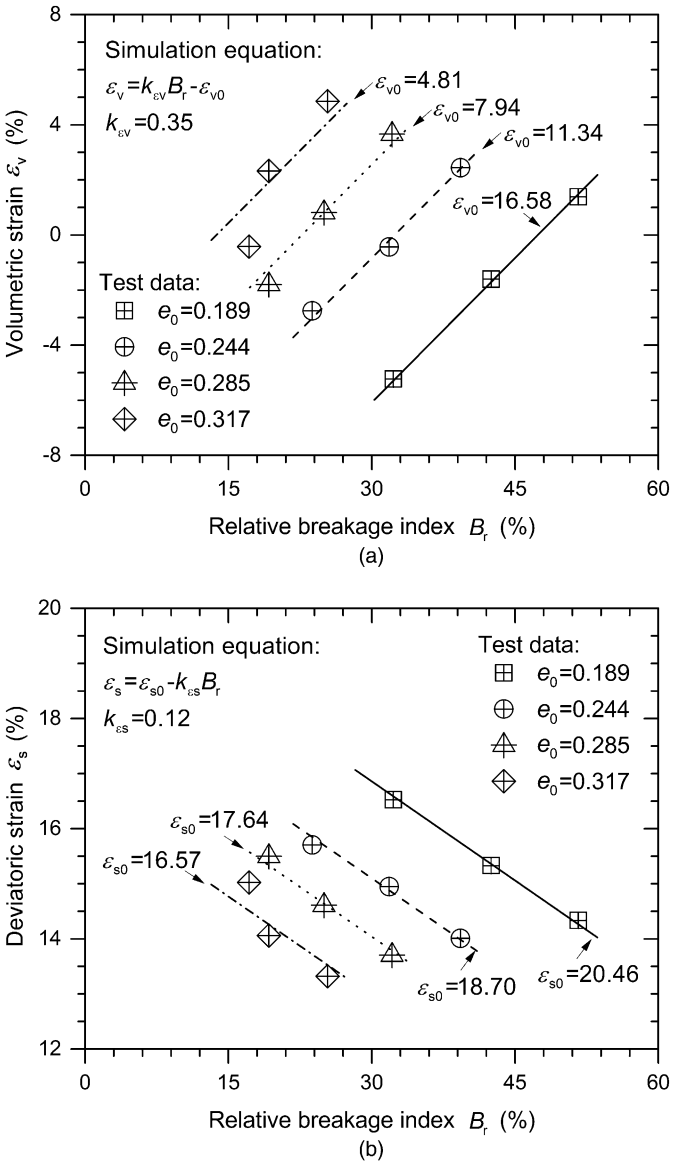


Fig. 10. Influence of the relative breakage index of TRM: (a) volumetric strain; (b) deviatoric strain

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volumetric strain ε_v of TRM at a given void ratio increases with an increase in the relative breakage index B_r, whereas the deviatoric strain ε_s decreases with an increase in the relative breakage index B_r. The influence of the relative breakage index B_r on the volumetric strain ε_v and deviatoric strain ε_s can be linearly expressed as

$$\varepsilon_v = k_{\varepsilon v} B_r - \varepsilon_{v0} \tag{14a}$$

$$\varepsilon_s = \varepsilon_{s0} - k_{\varepsilon s} B_r \tag{14b}$$

where $k_{\varepsilon v}$, ε_{v0}, $k_{\varepsilon s}$, and ε_{s0} = material parameters, which can be determined by a least-squares analysis. Table 3 lists the values of these material parameters (i.e., $k_{\varepsilon v}$, ε_{v0}, $k_{\varepsilon s}$, ε_{s0}). Eq. (14), as indicated in Fig. 10, can provide accurate estimations of the test results of TRM, except for when $e_0 = 0.317$. ε_{v0} and ε_{s0} are correlated with the initial void ratio e_0, as shown in Fig. 11. The expressions of ε_{v0} and ε_{s0} based on the initial void ratio e_0 can be expressed as

$$\varepsilon_{v0} = \varepsilon_{ev0} - \chi_{ev0} e_0 \tag{15a}$$

$$\varepsilon_{s0} = \varepsilon_{es0} - \chi_{es0} e_0 \tag{15b}$$

where χ_{ev0}, ε_{ev0}, χ_{es0}, and ε_{es0} = material parameters, which can be determined by a least-squares analysis. Table 3 provides the values of these material parameters (i.e., χ_{ev0}, ε_{ev0}, χ_{es0}, ε_{es0}). The test data of ε_{v0} and ε_{s0} based on the initial void ratio e_0 can be accurately expressed by Eq. (15). Substituting Eq. (15) into Eq. (14) yields

$$\varepsilon_v = \chi_{ev0} e_0 - \varepsilon_{ev0} + k_{\varepsilon v} B_r \tag{16a}$$

$$\varepsilon_s = \varepsilon_{es0} - \chi_{es0} e_0 - k_{\varepsilon s} B_r \tag{16b}$$

Eq. (16) indicates that the volumetric strain ε_v and deviatoric strain ε_s are related to the initial void ratio e_0 and relative breakage index B_r. As shown in Fig. 12, the residual volumetric strain ε_{rv} and residual deviatoric strain ε_{rs} can be expressed as

$$\varepsilon_{rv} = \varepsilon_v + \varepsilon_{v0} = k_{\varepsilon v} B_r \tag{17a}$$

$$\varepsilon_{rs} = \varepsilon_{s0} - \varepsilon_s = k_{\varepsilon s} B_r \tag{17b}$$

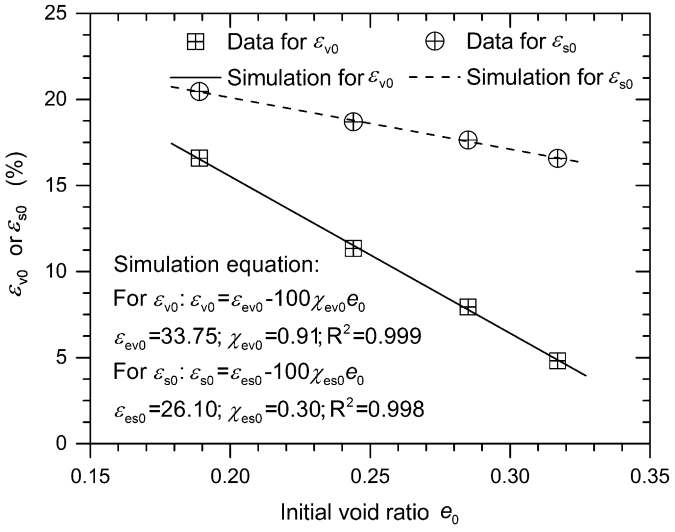


Fig. 11. Relationship between ε_{v0}, ε_{s0}, and e_0 for TRM

As shown in Fig. 12, the predictions by Eq. (17) correspond well with the test data for the residual volumetric strain ε_{rv} and residual deviatoric strain ε_{rs} based on the relative breakage index B_r.

The critical state void ratio e_{cs} is linearly related to the relative breakage index B_r, as shown in Fig. 13. The relationship between e_{cs} and B_r can be expressed as

$$e_{cs} = e_{B0} - k_e B_r / 100 \tag{18}$$

where e_{B0} and k_e = material parameters, which can be determined by a least-squares analysis. Table 3 lists the values of the two material parameters (i.e., e_{B0} and k_e). As shown in Fig. 13, the predictions by Eq. (18) are in good agreement with the test results in terms of the relationship between the critical state void ratio e_{cs} and relative breakage index B_r. In addition, the relationship between e_{cs} and B_r is independent of the initial void ratio e_0.

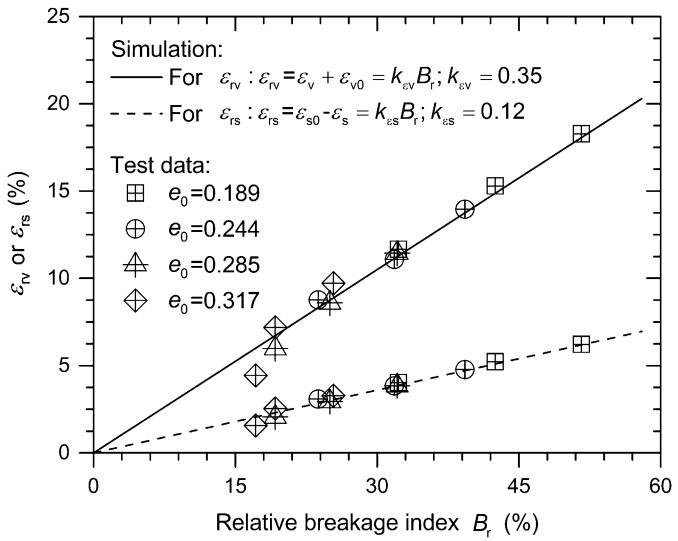


Fig. 12. Relationship between ε_{rv}, ε_{rs}, and B_r for TRM

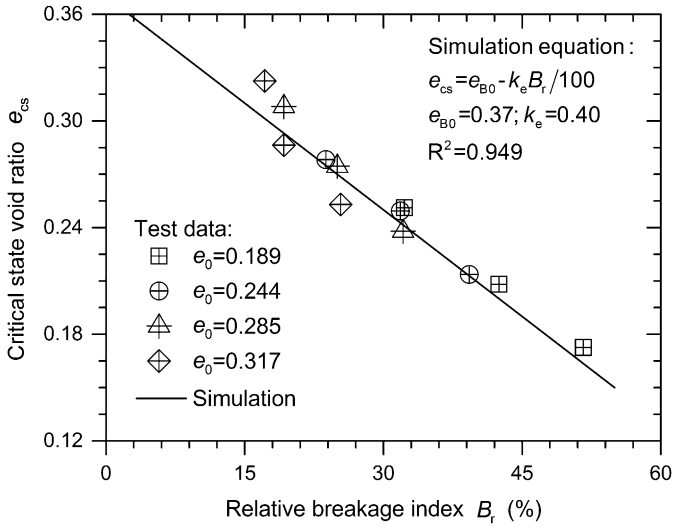


Fig. 13. Influence of the relative breakage index on the critical state void ratio for TRM

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Conclusions

This study investigated the relationships between the relative breakage index and mechanical indexes (e.g., peak state friction angle, critical state friction angle, initial elastic modulus, secant modulus at 50% of the peak strength, volumetric strain, deviatoric strain, critical state void ratio) based on a series of large-scale triaxial tests of TRM. The main conclusions can be summarized as follows.

1. Einav's relative breakage index (Einav 2007a) was adopted to quantify the degree of particle crushing of TRM during the shearing process. The fractal dimension of TRM at the end of the tests was found to be linearly correlated with the void ratio and logarithm of the confining pressure.
2. An increase in the relative breakage index resulted in an increase in the volumetric strain and decreases in the peak state friction angle, critical state friction angle, deviatoric strain, and critical state void ratio. Simple linear formulations were proposed for the peak state friction angle, critical state friction angle, initial elastic modulus, secant modulus at 50% of the peak strength, volumetric strain, deviatoric strain, and critical state void ratio based on the relative breakage index.

These conclusions could be useful for technologists to understand the deformation and strength of rockfill materials influenced by the particle breakage.

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Notation

The following symbols are used in this paper:

B_p = breakage potential;
B_r = relative breakage index;
B_t = total breakage potential;
d = diameter;
d_M = maximum diameter;
d_m = minimum diameter of the sieving size;
E_{0i} = initial elastic modulus;
E_{50} = secant modulus at 50% of the peak strength;
e_{B0} and k_e = material parameters in relation to e_{cs};
e_{cs} = critical state void ratio;
e_0 = initial void ratio;
$F_0(d)$, $F_c(d)$, and $F_u(d)$ = formulations of the IG, CG, and UG, respectively;
k_{E0} and E_{B0} = material parameters in relation to E_{0i};
k_{E50} and E_{B50} = material parameters in relation to E_{50};
$k_{\varepsilon s}$, ε_{s0}, χ_{es0}, and ε_{es0} = material parameters in relation to ε_s;
$k_{\varepsilon v}$, ε_{v0}, χ_{ev0}, and ε_{ev0} = material parameters in relation to ε_v;
p_a = atmospheric pressure;
p_0 = initial confining pressure;
α_c = current fractal dimension;
α_{e0}, χ_α, and k_α = material parameters in relation to α_c;
α_u = ultimate fractal dimension;
α_0 = initial fractal dimension;
ε_{rs} = residual deviatoric strain;
ε_{rv} = residual volumetric strain;
ε_s = deviatoric strain;
ε_v = volumetric strain;
ϕ_{cs} = critical state friction angle;
ϕ_{cs0}, k_{cs}, ϕ_{ecs}, and χ_{ecs} = material parameters in relation to ϕ_{cs};
ϕ_{ps} = peak state friction angle;
ϕ_{ps0}, k_{ps}, ϕ_{eps}, and χ_{eps} = material parameters in relation to ϕ_{ps};
ϕ_{rcs} = increment in the critical state friction angle; and
ϕ_{rps} = increment in the peak state friction angle.

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